WortUhr LED–Board rev4: Blockschaltbild

Eine Platine, viermal bestellt. Jede trägt das LED–Feld; die Master–Platine trägt zusätzlich Versorgung und ESP32.
Versorgungsnetze und blattübergreifende Signale sind global und in den Blättern als Versorgungssymbol oder globales Label gezeichnet.



Generiert mit tools/gen_sch.py
Versorgung: USB–C PD 15 V, Wandler 5 V / 6 A

Sheet: /
File: WortUhr.kicad_sch

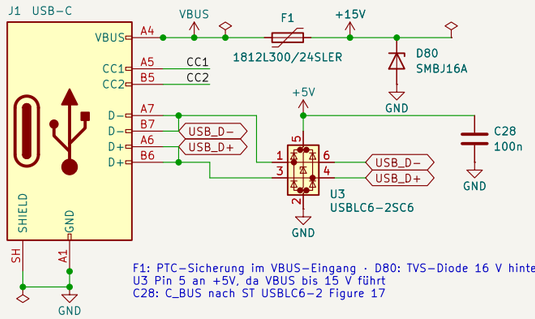
Title: WortUhr LED–Board (4x identisch, Master/Slave)

Size: A4	Date: 2026–11–10	**Rev: 4**
KiCad E.D.A. 10.0.6		Id: 1/4

Versorgung: USB-C Power Delivery 15 V, Abwärtswandler 5 V / 6 A, LDO 3,3 V

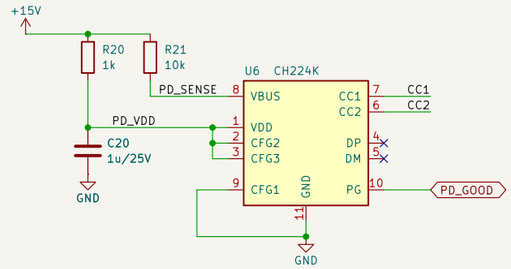
Alle Bauteile dieses Blatts werden nur auf der Master-Platine bestückt (Ausnahme: C1 sitzt auf jeder Platine).

USB-C Eingang, Sicherung, ESD-Schutz



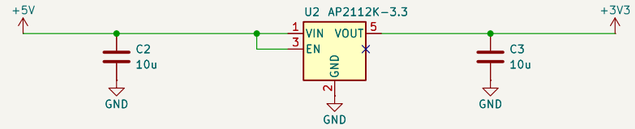
F1: PTC-Sicherung im VBUS-Eingang · D80: TVS-Diode 16 V hinter F1
U3 Pin 5 an +5V, da VBUS bis 15 V führt
C28: C_BUS nach ST USBLC6-2 Figure 17

USB-PD Sink: fordert 15 V an



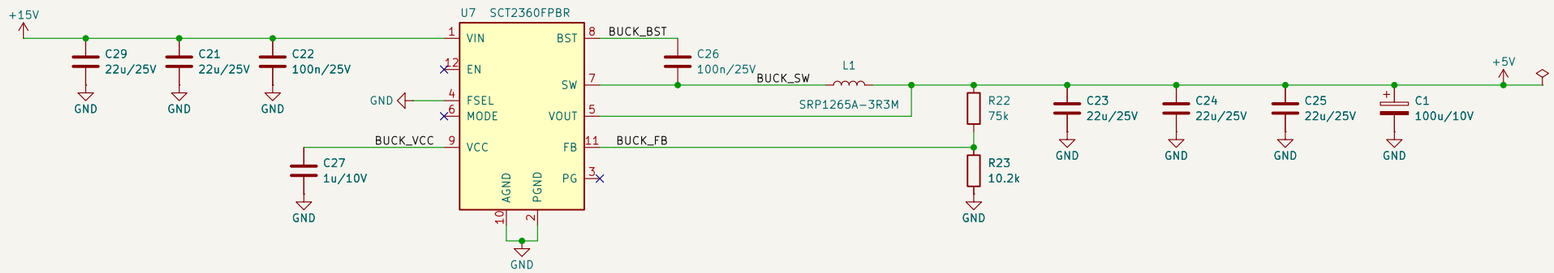
CFG1 = GND, CFG2 = CFG3 = VDD: Anforderung 15 V
R20/C20: Versorgung VDD aus +15V, R21: Spannungsmessung
PG: Open-Drain, low = PD-Spannung liegt an (Pull-up im ESP32)

LDO 3,3 V für ESP32 und Sensor



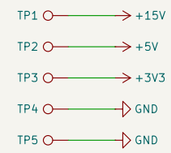
EN an VIN: Regler dauernd eingeschaltet
C2/C3: Stabilität nach Datenblatt (Regel PWR-003)

Abwärtswandler 15 V → 5 V / 6 A



FSEL = GND: 400 kHz · MODE offen: USM · EN offen: Start automatisch
R22/R23: Vout = Vref × (1 + 75k / 10,2k); mit Vref = 0,6 V ergibt das 5,01 V
C21/C29/C22 direkt an VIN und PGND (Hot-Loop, SCT2360 Figure 22; zwei 22 µF laut Datenblatt) · C1: Pufferelko, auf jeder Platine

Testpunkte (Pads Unterseite)



Energiekette: USB-C PD 15 V → F1 → U7 (5 V, bis 6 A) → LED-Feld und U2 (3,3 V).
Die Firmware begrenzt den LED-Strom auf geschätzte 5 A und schaltet die LEDs nur bei PD_GOOD ein.

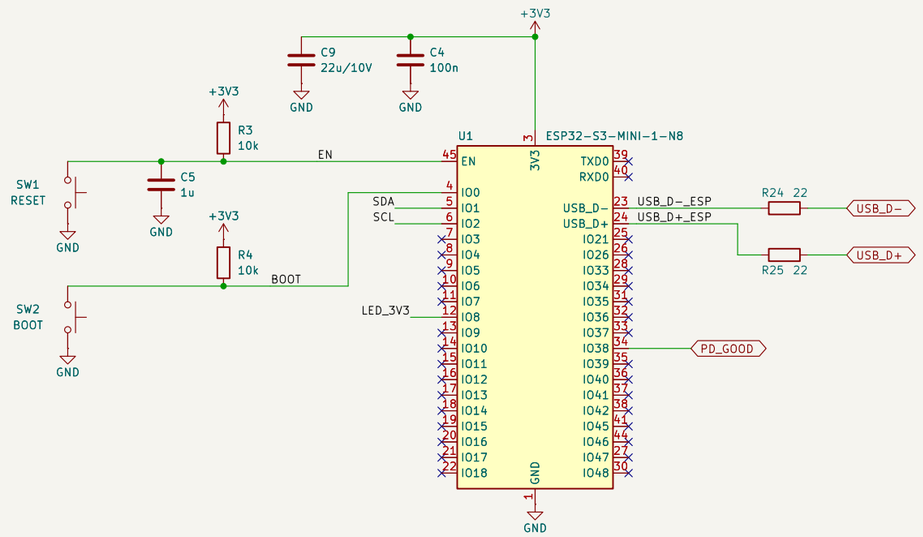
Generiert mit tools/gen_sch.py

Sheet: /Versorgung/		
File: Versorgung.kicad_sch		
Title: Versorgung: USB-C PD, Wandler, LDO		
Size: A3	Date: 2026-11-10	**Rev: 4**
KiCad E.D.A. 10.0.6		Id: 2/4

ESP32-S3, Pegelwandler für die LED-Daten, Umgebungslichtsensor

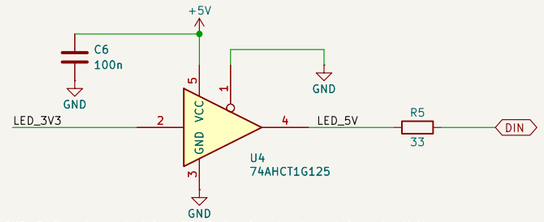
Alle Bauteile dieses Blatts werden nur auf der Master-Platine bestückt.

ESP32-S3-MINI-1 mit Reset, Boot und USB



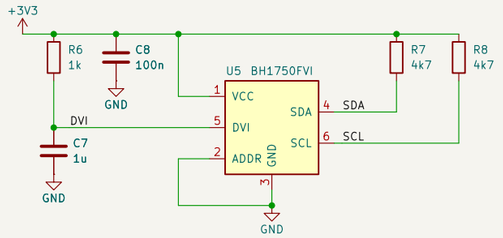
USB: natives USB-Serial-JTAG, R24/R25 nahe am Modul (Espressif)
IO0 = BOOT (SW2), auch Bedientaster der Firmware
IO38 = PD_GOOD mit internem Pull-up · C4/C9 direkt am 3V3-Pin

Pegelwandler LED-Daten 3,3 V → 5 V



OE = GND: Treiber dauernd aktiv · R5: Serienwiderstand am Anfang der Kette

Umgebungslichtsensor BH1750 (I2C 0x23)



ADDR = GND: Adresse 0x23 · DVI: Reset über R6/C7 beim Einschalten
R7/R8: I2C-Pull-ups · Sensor sitzt unter dem Lichtleiter

Testpunkte (Pads Unterseite)



Generiert mit tools/gen_sch.py

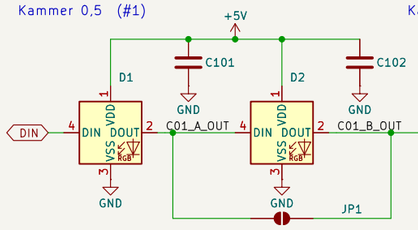
Sheet: /ESP32/
File: ESP32.kicad_sch

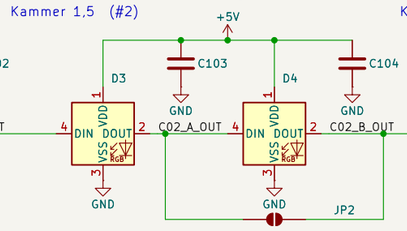
Title: ESP32-S3, Pegelwandler, Lichtsensor

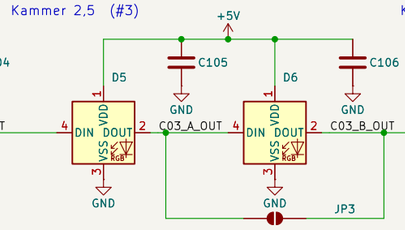
Size: A3	Date: 2026-11-10	**Rev: 4**
KiCad E.D.A. 10.0.6		Id: 3/4

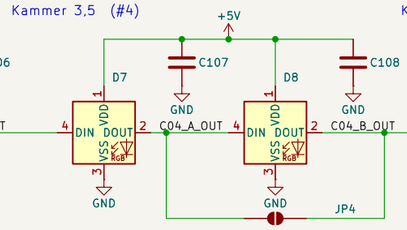
LED-Feld: 36 Kammern in der Reihenfolge der Datenkette (alle Platinen)

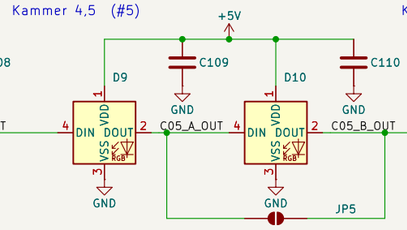
Kammer x,y = Spalte,Zeile auf der Master-Platine · D1…D75: SK6812RGBW · C101…C175: 100 nF am +5V-Pin der LED
LED A wird immer bestückt, LED B wahlweise · JPn schließen, wenn LED B fehlt

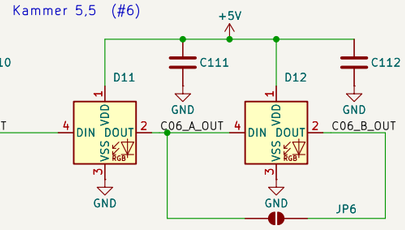
Kammer 0,5 (#1)


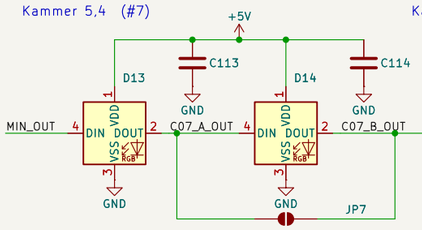
Kammer 1,5 (#2)


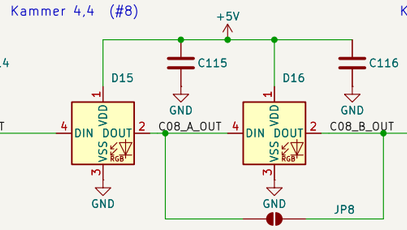
Kammer 2,5 (#3)


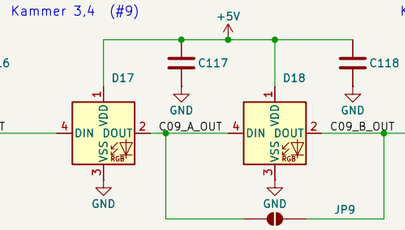
Kammer 3,5 (#4)


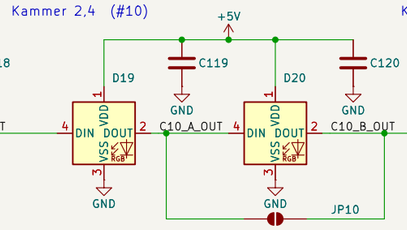
Kammer 4,5 (#5)


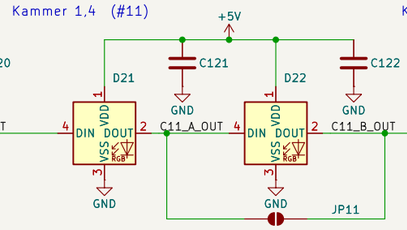
Kammer 5,5 (#6)


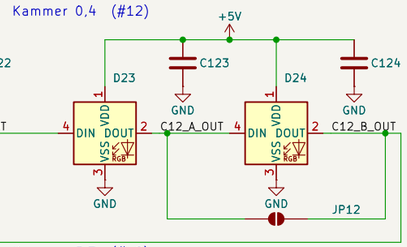
Kammer 5,4 (#7)


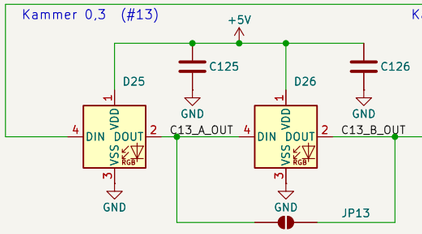
Kammer 4,4 (#8)


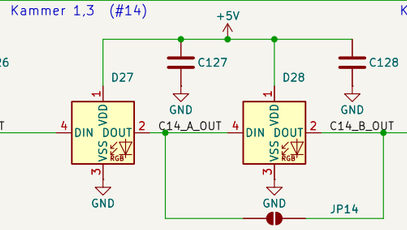
Kammer 3,4 (#9)


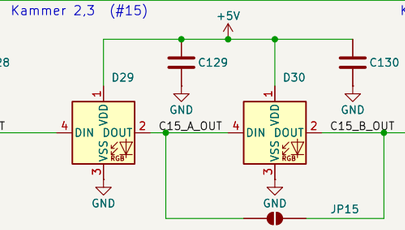
Kammer 2,4 (#10)


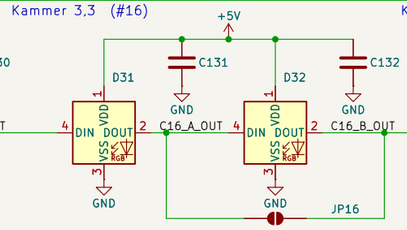
Kammer 1,4 (#11)


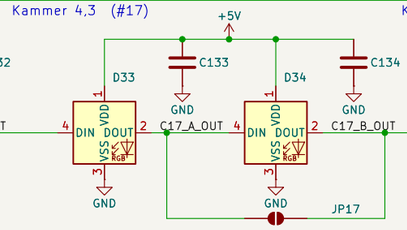
Kammer 0,4 (#12)


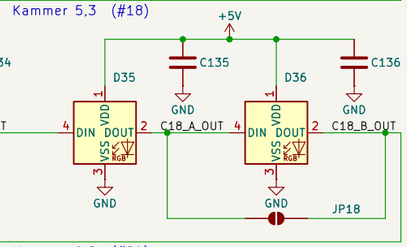
Kammer 0,3 (#13)


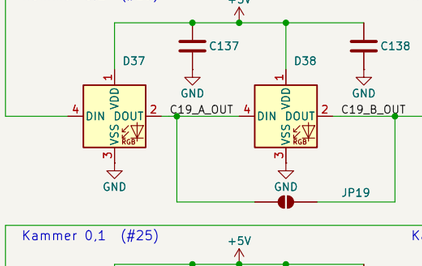
Kammer 1,3 (#14)


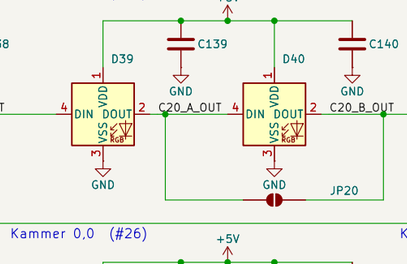
Kammer 2,3 (#15)


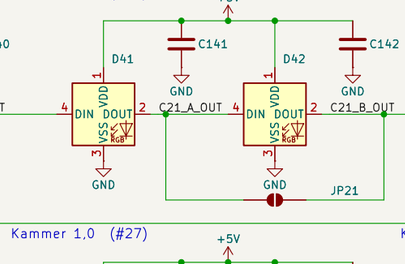
Kammer 3,3 (#16)


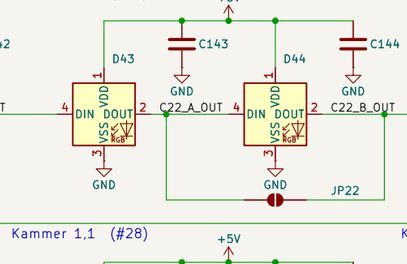
Kammer 4,3 (#17)


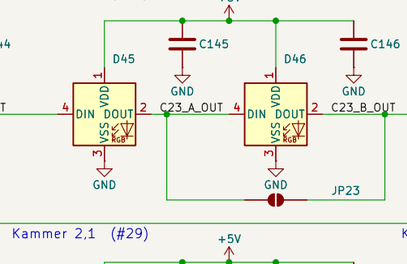
Kammer 5,3 (#18)


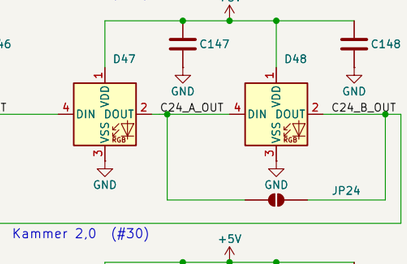
Kammer 5,2 (#19)


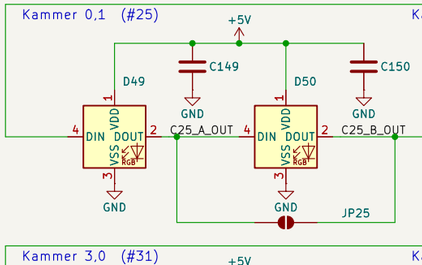
Kammer 4,2 (#20)


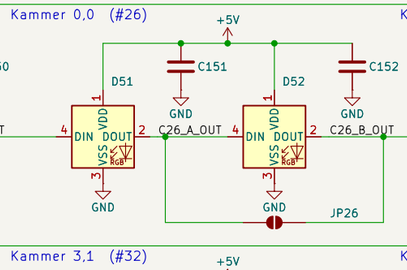
Kammer 3,2 (#21)


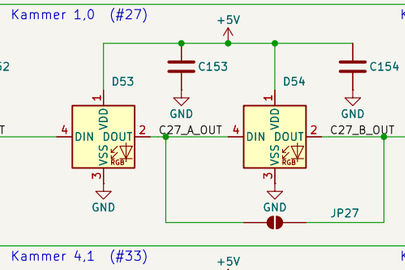
Kammer 2,2 (#22)


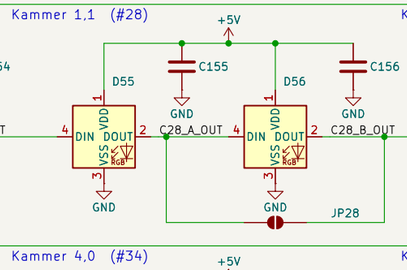
Kammer 1,2 (#23)


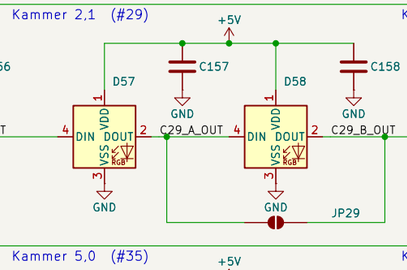
Kammer 0,2 (#24)


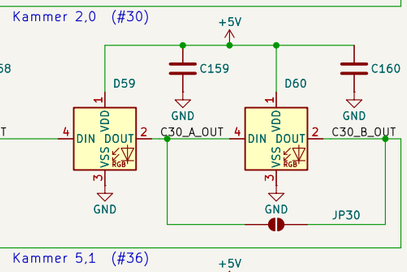
Kammer 0,1 (#25)


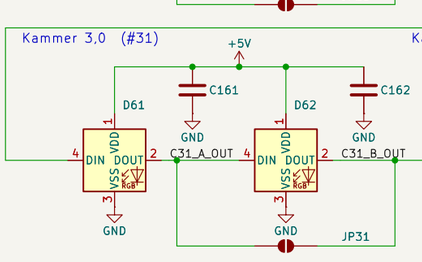
Kammer 0,0 (#26)


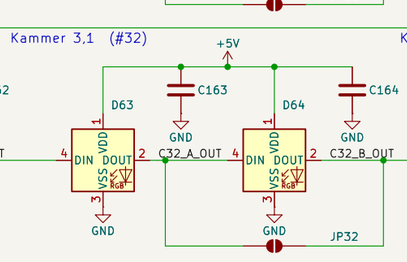
Kammer 1,0 (#27)


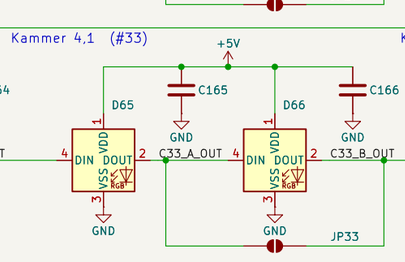
Kammer 1,1 (#28)


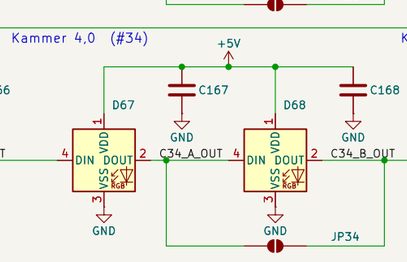
Kammer 2,1 (#29)


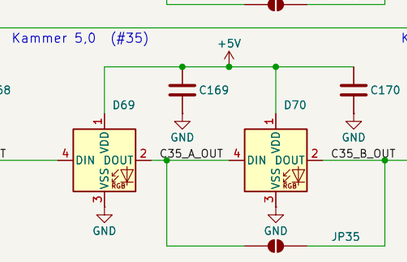
Kammer 2,0 (#30)


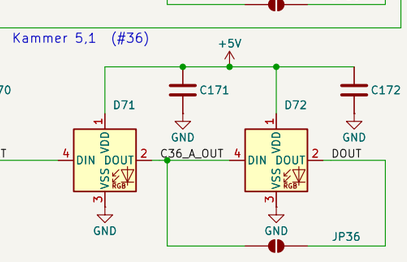
Kammer 3,0 (#31)


Kammer 3,1 (#32)


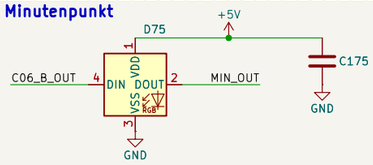
Kammer 4,1 (#33)


Kammer 4,0 (#34)


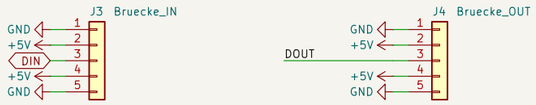
Kammer 5,0 (#35)


Kammer 5,1 (#36)


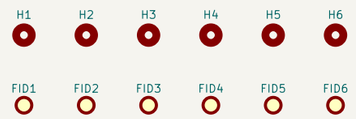
Minutenpunkt



Kantenbrücken zur Nachbarplatine (OR 2512, von Hand)



Montagelöcher M3, Passermarken



Generiert mit tools/gen_sch.py

Sheet: /LED_Feld/
File: LED_Feld.kicad_sch

Title: LED-Feld, Abblockung und Verbinder

Size: A3	Date: 2026-11-10	**Rev: 4**
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